

A Three-Dimensional Keller Map with a Rational Collision

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Abstract

We construct a noninjective étale endomorphism of affine three-space from a marked-divisor cover of $\mathbb{P}^1$. An auxiliary conic and residual intersection identify its source with affine three-space without coordinates. We then give an explicit polynomial realization, exact algebraic certificates, the image and fiber structure over the complex numbers, and a normalized weighted marked-root family with symmetric monodromy.

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1 The geometric counterexample

Let k be a field of characteristic zero. Consider the finite flat degree-three marking morphism

$$\pi : \mathbb{P}^1 \times \mathrm{Sym}^2(\mathbb{P}^1) \longrightarrow \mathrm{Sym}^3(\mathbb{P}^1), \quad (x, Q) \longmapsto x + Q.$$

Its ramification divisor is

$$D = \{(x, Q) : Q \text{ contains } x\}.$$

Choose $0, \infty \in \mathbb{P}^1$ and let $H \subset \mathrm{Sym}^3(\mathbb{P}^1)$ be the hyperplane whose equation on the affine line $\mathbb{P}^1 \setminus \{\infty\}$ is

$$x + y + z = 0. \tag{1}$$

On the small diagonal this becomes $3x = 0$; after projective closure the remaining intersection multiplicity two lies at ∞ . Thus H is tangent but not osculating to the twisted cubic of triple points.

Set

$$E = \pi^{-1}(H), \quad U = (\mathbb{P}^1 \times \mathrm{Sym}^2(\mathbb{P}^1)) \setminus (D \cup E), \quad Y = \mathrm{Sym}^3(\mathbb{P}^1) \setminus H.$$

Since H is a hyperplane in $\mathbb{P}^3 \simeq \mathrm{Sym}^3(\mathbb{P}^1)$, $Y \simeq \mathbb{A}^3$.

1.1 The auxiliary conic

For fixed x , let D_x and E_x be the lines cut out by D and E in $\{x\} \times \mathrm{Sym}^2(\mathbb{P}^1)$. If $\Gamma = \{2t : t \in \mathbb{P}^1\}$ is the Veronese conic, then D_x is tangent to Γ at $2x$, while the E_x form the pencil through $p_\infty = 2\infty$. Define

$$\tau(x) = -x/2, \quad \alpha = \tau^{-1}, \quad h(x) = D_x \cap E_x = x + \alpha(x), \quad \alpha(x) = -2x,$$

where the last plus sign denotes addition of effective divisors. The image $C = h(\mathbb{P}^1)$ is a smooth conic, and the boundary lines become its chords

$$E_x = \overline{h(x)p_\infty}, \quad D_x = \overline{h(x)h(\tau(x))}, \tag{2}$$

with coincident endpoints interpreted as tangency [Cha26].

Proposition 1.1 (affine source). *The residual-intersection construction gives $U \simeq \mathbb{A}^3$.*

Proof. For $(x, Q) \in U$, the line $\overline{h(x)Q}$ has a second intersection

$$\rho(x, Q) \in C.$$

This varies regularly, with tangency counted twice. By (2), the value p_∞ occurs exactly on the removed divisor E , so ρ takes values in $C \setminus \{p_\infty\} \simeq \mathbb{A}^1$.

For a residual point s , the removed divisor D excludes exactly the one marked point satisfying $s = h(\tau(x))$. Hence

$$B = \{(x, s) \in \mathbb{P}^1 \times (C \setminus \{p_\infty\}) : s \neq h(\tau(x))\} \longrightarrow C \setminus \{p_\infty\}$$

is the complement of a section in a $\mathbb{P}^1$ -bundle. It is a torsor under a line bundle; both the line bundle and torsor are trivial on $\mathbb{A}^1$, so $B \simeq \mathbb{A}^2$.

For fixed $(x, s) \in B$, the point Q varies on $\overline{h(x)s} \setminus \{h(x)\} \simeq \mathbb{A}^1$. This is again the complement of a section in a $\mathbb{P}^1$ -bundle. Its associated line bundle and torsor are trivial on $B \simeq \mathbb{A}^2$. Therefore

$$U \longrightarrow B \longrightarrow \mathbb{A}^1, \quad U \simeq \mathbb{A}^3. \quad \square$$

Theorem 1.2 (geometric counterexample). *After choosing affine-space identifications, the restriction of π gives an étale, generically three-to-one morphism*

$$\mathbb{A}^3 \simeq U \longrightarrow Y \simeq \mathbb{A}^3.$$

Its Jacobian determinant in these coordinates is a nonzero constant. In particular it is not an isomorphism and, after base change to an algebraic closure, is not injective.

Proof. The map π is the finite flat cover that marks one point of an effective divisor of degree three. Its ramification divisor is D , so its restriction to U is étale. A divisor with three distinct points has three markings; hence the restriction has generic degree three. Proposition 1.1 and the identification $Y \simeq \mathbb{A}^3$ give the displayed affine morphism. Its Jacobian determinant is a unit in a polynomial ring and is therefore a nonzero constant. $\square$

This proves the existence of the counterexample without choosing polynomial coordinates. The next two sections give a compact explicit formula and then certify, including at the projective root at infinity, that it is a coordinate realization of this geometric morphism.

2 The normalized factorization bridge

The geometric construction admits a short affine realization before the expanded polynomial is written down. Let

$$L = aT + bS, \quad Q = cT^2 + dTS + eS^2.$$

Then

$$LQ = (ac)T^3 + (ad + bc)T^2S + (ae + bd)TS^2 + (be)S^3$$

and

$$\text{Res}(L, Q) = a^2e - abd + b^2c.$$

Proposition 2.1 (coefficient–resultant coordinates). *The morphism*

$$\Theta : \mathbb{A}^2 \times \mathbb{A}^3 \longrightarrow \mathbb{A}^4 \times \mathbb{A}^1, \quad (L, Q) \longmapsto (LQ, \text{Res}(L, Q))$$

is étale on the nonzero-resultant locus.

Proof. Suppose that a tangent vector $(\dot{L}, \dot{Q})$ is killed by the differential of multiplication. Then

$$\dot{L}Q + L\dot{Q} = 0.$$

When the resultant is nonzero, L and Q are coprime. Hence L divides $\dot{L}$, and degree considerations give

$$(\dot{L}, \dot{Q}) = \lambda(L, -Q).$$

This is the infinitesimal relative-scaling direction. The resultant has bidegree $(2, 1)$, so

$$d\text{Res}_{(L, Q)}(L, -Q) = (2 - 1)\text{Res}(L, Q) \neq 0.$$

Thus $d\Theta$ has trivial kernel; source and target both have dimension five. In the displayed coordinates the same statement is

$$\det D\Theta = -\text{Res}(L, Q)^2.$$

$\square$

Consider the normalized slice

$$X_{\text{fac}} = \{(a, b, c, d, e) : a^2e - abd + b^2c = 1, \quad ad + bc = 1\}.$$

The multiplication map

$$\mu : X_{\text{fac}} \longrightarrow \mathbb{A}^3, \quad (a, b, c, d, e) \longmapsto (ac, ae + bd, be) \quad (3)$$

is the base change of Θ to the slice on which the resultant and the T^2S coefficient are both one. It is therefore etale, without a separate calculation at $a = 0$.

Let

$$U = (\mathbb{P}(V_1) \times \mathbb{P}(V_2)) \setminus (\{\text{Res} = 0\} \cup \{[LQ]_{T^2S} = 0\}).$$

This is the projective source open from Section 1.

Lemma 2.2 (unique affine normalization). *Projectivization induces an isomorphism*

$$X_{\text{fac}} \xrightarrow{\sim} U.$$

Proof. Choose representatives of $([L], [Q]) \in U$ and put

$$m = [LQ]_{T^2S}, \quad r = \text{Res}(L, Q).$$

Under $(L, Q) \mapsto (\lambda L, \mu Q)$ these functions have weights $(1, 1)$ and $(2, 1)$, respectively. The unique solution of

$$\lambda \mu m = 1, \quad \lambda^2 \mu r = 1$$

is

$$\lambda = \frac{m}{r}, \quad \mu = \frac{r}{m^2}.$$

If the original representatives are replaced by $(\alpha L, \beta Q)$, the new normalization scalars are λ/α and μ/β . Hence the normalized pair is independent of representatives. The formulas are regular on U and invert projectivization. $\square$

Proposition 2.3 (polynomial slice coordinates). *The morphism $\Phi : \mathbb{A}_{a,y,z}^3 \rightarrow X_{\text{fac}}$ defined by*

$$\begin{aligned} b &= 1 + ay, \\ c &= 1 - \frac{3}{2}ay + a^2z, \\ d &= \frac{1}{2}y - az + \frac{3}{2}ay^2 - a^2yz, \\ e &= -2z + 4y^2 - 4ayz + 3ay^3 - 2a^2y^2z \end{aligned}$$

is an isomorphism. Its inverse is

$$y = 2bd - ae, \quad z = 2d^2 + ce + 6bd^2 + 3bce - \frac{9}{2}e. \quad (4)$$

Proof. Substitution of the forward formulas gives

$$ad + bc = 1, \quad a^2e - abd + b^2c = 1.$$

Substituting them into (4) returns y, z . Conversely, substitution of (4) into the forward formulas returns b, c, d, e modulo the two defining equations of X_{fac} . All formulas are polynomial, including on the divisor $a = 0$. $\square$

The residual torus action

$$\lambda \cdot (a, b, c, d, e) = (\lambda a, b, c, \lambda^{-1}d, \lambda^{-2}e)$$

becomes

$$\lambda \cdot (a, y, z) = (\lambda a, \lambda^{-1}y, \lambda^{-2}z).$$

Thus the weights $(1, -1, -2)$ are the relative factor-scaling weights on the normalized slice.

Put

$$G = \mu \circ \Phi : \mathbb{A}_{a,y,z}^3 \longrightarrow \mathbb{A}^3.$$

Exact expansion gives

$$\det DG = -1.$$

The polynomial F of Section 3 is obtained from G by linear source and target changes. Namely, with

$$A(z_1, z_2, z_3) = \left(z_1, z_2, -\frac{1}{2}z_3 \right), \quad B(u_1, u_2, u_3) = (u_3, 2u_2, 2u_1),$$

one has

$$F = B \circ G \circ A. \tag{5}$$

Consequently

$$\det DF = (\det B)(\det DG)(\det A) = -2.$$

This separates the three mechanisms. Etaleness comes from coprimality and the nonzero resultant weight; generic degree three comes from the three choices of linear factor of a squarefree cubic; and the exceptional geometric input is Proposition 2.3, identifying this tangent nonosculating slice with affine three-space.

2.1 Unequal factor degrees

The differential argument extends uniformly. Put

$$V_p = \text{Sym}^p(k^2), \quad V_q = \text{Sym}^q(k^2)$$

and define

$$\Theta_{p,q} : V_p \times V_q \longrightarrow V_{p+q} \times \mathbb{A}^1, \quad (A, B) \longmapsto (AB, \text{Res}(A, B)).$$

Proposition 2.4 (unequal-degree coefficient–resultant theorem). *If $p \neq q$, then $\Theta_{p,q}$ is etale on the coprime locus.*

Proof. The kernel of the differential of multiplication at a coprime pair is again spanned by $(A, -B)$. Since the resultant has bidegree (q, p) ,

$$d\text{Res}_{(A,B)}(A, -B) = (q - p)\text{Res}(A, B).$$

This is nonzero in characteristic zero when $p \neq q$, and the source and target dimensions are both $p + q + 2$. $\square$

There is a second distinction at the normalization stage. Suppose $q > p$, let m be a nonzero linear functional of AB , and put $r = \text{Res}(A, B)$. The conditions

$$\lambda \mu m = 1, \quad \lambda^q \mu^p r = 1$$

are equivalent to

$$\lambda^{q-p} = \frac{m^p}{r}, \quad \mu = \frac{1}{\lambda m}.$$

For consecutive degrees $q - p = 1$, this gives the unique normalization

$$\lambda = \frac{m^p}{r}, \quad \mu = \frac{r}{m^{p+1}}.$$

For $q - p > 1$ there remains a residual μ_{q-p} ambiguity. When $p = q$, the resultant has weight zero along relative scaling and does not repair the multiplication kernel. Thus unequal degrees explain etaleness, while consecutive degrees explain the canonical normalized slice used above.

3 The explicit Keller map

Throughout Sections 3–5, let k be a field of characteristic zero. Write $\mathbb{A}_k^m$ for affine m -space over k . Set

$$u = 1 + xy$$

and define $F = (F_1, F_2, F_3) : \mathbb{A}_k^3 \rightarrow \mathbb{A}_k^3$ by

$$\begin{aligned} F_1 &= u^3 z + y^2 u(4 + 3xy), \\ F_2 &= y + 3xu^2 z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 z. \end{aligned} \tag{6}$$

Theorem 3.1 (constant Jacobian and collision). *The morphism F is etale and*

$$\det DF = -2.$$

Moreover the three distinct k -points

$$p_0 = (0, 0, -1/4), \quad p_+ = (1, -3/2, 13/2), \quad p_- = (-1, 3/2, 13/2)$$

have the common image $(-1/4, 0, 0)$. Hence F is not injective and is not a polynomial automorphism. For every $d \geq 3$, the product $F \times \text{id}_{\mathbb{A}^{d-3}}$ has the same properties.

Proof. On the dense open subset $D(x) \subset \mathbb{A}_k^3$, put

$$t = y + x^{-1}, \quad r = 2x^{-1}, \quad c = F_3(x, y, z),$$

and write $(a, b, c) = F(x, y, z)$. Direct expansion in $k[x, x^{-1}, y, z]$ gives

$$a = t^2 + \frac{rt}{2} - ct^3, \quad b = r + 4t - 3ct^2. \tag{7}$$

Consequently

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}, \quad \det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x.$$

Since $r = 2/x$, the chain rule gives $\det DF = -2$ on $D(x)$. The determinant is a polynomial, and $D(x)$ is dense because k is an infinite field, so the identity holds on all of $\mathbb{A}_k^3$. Since -2 is a unit, the Jacobian criterion makes F etale.

Substitution into (6) gives

$$F(p_0) = F(p_+) = F(p_-) = (-1/4, 0, 0).$$

The three first coordinates $0, 1, -1$ show that the source points are distinct. The final assertion follows because the determinant of a product map is the product of the determinants and the displayed collision persists after fixing the additional coordinates. $\square$

The proof above is entirely algebraic and applies over every characteristic-zero field. Independent exact implementations, including a separately authored Lean formalization, verify the determinant and collision; they are verification artifacts rather than inputs to the argument.

4 Global coordinate certificate

We now identify the geometric morphism of Theorem 1.2 with the polynomial map (6). The only genuinely global coordinate issue is the marked root at infinity; the two root charts below isolate that check.

Let (a, b, c) be coordinates on $Y = \mathbb{A}_k^3$ and define

$$Q_{a,b,c}(U, V) = cU^3 - 2U^2V + bUV^2 - 2aV^3. \quad (8)$$

The *raw root incidence* is the closed subscheme

$$I = V(Q_{a,b,c}) \subset Y \times \mathbb{P}_k^1.$$

Because the coefficient of U^2V is the unit -2 , no fiber contains the whole projective line. Each fiber is therefore an effective Cartier divisor of degree three on $\mathbb{P}_k^1$, and $I \rightarrow Y$ is finite flat of degree three. Let $I^{\text{simp}} \subset I$ be the open locus where the marked projective root has multiplicity one. No normalization is used in this cubic model.

Proposition 4.1 (source marking). *The formula*

$$\Phi : \mathbb{A}_k^3 \longrightarrow I, qquad(x, y, z) \longmapsto (F(x, y, z), [1 + xy : x])$$

defines a morphism, and its image is contained in I^{simp} .

Proof. Expansion of (8) after substituting (6) gives the polynomial identity

$$Q_{F(x,y,z)}(1 + xy, x) = 0.$$

The homogeneous pair $(1 + xy, x)$ never vanishes simultaneously, so it defines a morphism to $\mathbb{P}_k^1$. Simplicity will follow on each of the two charts below; their source preimages are $D(x)$ and $D(1 + xy)$, which cover $\mathbb{A}_k^3$. $\square$

4.1 The chart $V \neq 0$

Put $u = U/V$ and

$$P(T) = cT^3 - 2T^2 + bT - 2a.$$

The simple-root chart has coordinate ring

$$k[a, b, c, u, P'(u)^{-1}]/(P(u)). \quad (9)$$

Define $v = P'(u)/2$ and

$$x = v^{-1}, \quad y = u - v, \quad qquad z = 5v^2 - 3uv - cv^3. \quad (10)$$

These are regular functions on (9). Conversely, on $D(x) \subset \mathbb{A}_k^3$ one has

$$u = y + x^{-1}, \quad v = x^{-1}, \quad qquad a = u^2 + uv - cu^3, \quad qquad b = 4u + 2v - 3cu^2, \quad (11)$$

and hence $P(u) = 0$ and $P'(u) = 2v = 2/x$.

Lemma 4.2. *The maps (10) and (11) are mutually inverse between $D(x)$ and the chart (9).*

Proof. Starting from $D(x)$, (11) gives $v = 1/x$, so (10) returns x and $y = u - v$. Solving the identity $c = 2x - 3x^2y - x^3z$ for z after substituting $x = 1/v$ and $y = u - v$ gives the third formula in (10); hence the source composition is the identity.

Conversely, start in (9). The equation $P(u) = 0$ is equivalent to $a = u^2 + uv - cu^3$, while $P'(u) = 2v$ is equivalent to $b = 4u + 2v - 3cu^2$. Substitution of (10) into (6) therefore returns (a, b, c) , and $y + 1/x = (u - v) + v = u$. Thus the incidence composition is also the identity. $\square$

4.2 The chart $U \neq 0$

Put $s = V/U$ and

$$R(s) = c - 2s + bs^2 - 2as^3, \text{quadd} = 1 - bs + 3as^2 = -\frac{1}{2}R'(s). \quad (12)$$

The simple-root chart has coordinate ring

$$k[a, b, c, s, d^{-1}]/(R(s)).$$

Define

$$x = s/d, \quad y = b - 3as, \quad (13)$$

$$z = d(a - 4b^2 + (b^3 + 22ab)s - (30a^2 + 8ab^2)s^2 + 21a^2bs^3 - 18a^3s^4). \quad (14)$$

The expression (14) is polynomial at $s = 0$. There the root equation gives $c = 0$, $d = 1$, and the inverse is

$$(x, y, z) = (0, b, a - 4b^2). \quad (15)$$

Lemma 4.3. *Equations (13)–(14) define a morphism to $\mathbb{A}_k^3$ inverse to Φ over the chart $U \neq 0$.*

Proof. For $s \neq 0$, solving first gives

$$z = \frac{2sd^2 - 3s^2d(b - 3as) - cd^3}{s^3}.$$

Using $R(s) = 0$, or $c = 2s - bs^2 + 2as^3$, its numerator is divisible by s^3 ; the quotient is exactly (14). Thus the displayed formula extends the inverse across $s = 0$.

In the coordinate ring $k[a, b, c, s, d^{-1}]/(R)$, direct substitution into (6) gives $(F_1, F_2, F_3) = (a, b, c)$ and

$$[1 + xy : x] = [1 : s].$$

For the reverse composition, on $D(1 + xy)$ put

$$s = \frac{x}{1 + xy}.$$

Substitution of $F(x, y, z)$ into (12) gives $R(s) = 0$ and $d = (1 + xy)^{-1}$. Equations (13)–(14) then return (x, y, z) . All identities are identities in the indicated localized polynomial rings, so the verification includes $s = 0$ and does not rely on taking a limit. $\square$

4.3 Agreement on the overlap

On $U, V \neq 0$,

$$u = s^{-1}, qquadv = d/s. \quad (16)$$

The first two coordinates in (10) become $x = s/d$ and $y = 1/s - d/s = b - 3as$, using (12) and $R(s) = 0$. Substituting (16) into the third affine formula and again using $R(s) = 0$ gives (14). Hence the two inverse maps agree scheme-theoretically on their overlap.

Theorem 4.4 (cubic marked-root isomorphism). *The morphism Φ induces an isomorphism*

$$\mathbb{A}_k^3 \xrightarrow{\sim} I^{\text{simp}}.$$

Under this isomorphism, F is the restriction to I^{simp} of the finite flat projection $I \rightarrow \mathbb{A}_k^3$.

Proof. The two root charts cover $\mathbb{P}^1$, and their inverse morphisms agree by (16). Their source preimages $D(x)$ and $D(1 + xy)$ cover $\mathbb{A}_k^3$. Lemmas 4.2 and 4.3 therefore glue to a two-sided inverse of Φ . $\square$

Under $\text{Sym}^3(\mathbb{P}^1) \simeq \mathbb{P}^3$, normalizing the complement of the hyperplane H from Section 1 makes the U^2V coefficient equal to -2 and identifies Y with $\mathbb{A}_k^3$. Marking one root gives I , and removing the repeated-marked-root divisor gives I^{simp} . Thus Theorem 4.4 is precisely the global coordinate certificate for Theorem 1.2.

5 Fibers, image, and nonproperness

For this section, assume first that k is algebraically closed of characteristic zero. Put

$$\Delta(a, b, c) = 27a^2c^2 - 18abc + 16a + b^3c - b^2 \quad (17)$$

and

$$\Gamma = V(3bc - 4, 12a - b^2) \subset \mathbb{A}_k^3. \quad (18)$$

The discriminant of $P(T) = cT^3 - 2T^2 + bT - 2a$ is -4Δ .

Proposition 5.1 (geometric fibers and image). *For every $q = (a, b, c) \in \mathbb{A}_k^3$,*

$$\#F^{-1}(q) = \begin{cases} 3, & \Delta(q) \neq 0, \\ 1, & \Delta(q) = 0 \text{ and } q \notin \Gamma, \\ 0, & q \in \Gamma. \end{cases}$$

Consequently $F(\mathbb{A}_k^3) = \mathbb{A}_k^3 \setminus \Gamma$.

Proof. By Theorem 4.4, the fiber is the set of simple projective roots of the nonzero binary cubic (8). A binary cubic over an algebraically closed field has multiplicity type $(1, 1, 1)$, $(2, 1)$, or (3) , yielding respectively three, one, or zero simple roots. The first type is equivalent to $\Delta \neq 0$. A comparison of coefficients with a cube shows that type (3) occurs exactly when $3bc = 4$ and $12a = b^2$; these equations force $c \neq 0$. This is precisely Γ , and all remaining discriminant points have type $(2, 1)$. $\square$

We now specialize to $k = \mathbb{C}$. For a polynomial map $f : \mathbb{C}^m \rightarrow \mathbb{C}^m$, define its nonproperness set S_f to be the set of targets q for which there are $p_j \in \mathbb{C}^m$ with $\|p_j\| \rightarrow \infty$ and $f(p_j) \rightarrow q$.

Proposition 5.2 (exact nonproperness set). *For the map F in (6),*

$$S_F = V(\Delta) \subset \mathbb{C}^3.$$

Proof. Let $q_0 = (a_0, b_0, c_0) \in V(\Delta)$. Its cubic has a repeated finite root t_0 : a repeated root cannot be the projective root $[1 : 0]$, since on the $s = V/U$ chart one has $R'(0) = -2$. Keep $b = b_0$, $c = c_0$, let $t \rightarrow t_0$, and set

$$a(t) = \frac{ct^3 - 2t^2 + bt}{2}.$$

Then $P_t(t) = 0$, the targets $(a(t), b, c)$ tend to q_0 , and $P'_t(t) = 3ct^2 - 4t + b$ tends to zero. Away from its finitely many zeros, the affine reconstruction (10) is defined and $x = 2/P'_t(t)$ is unbounded. Hence $V(\Delta) \subseteq S_F$.

Conversely, take $q_0 \notin V(\Delta)$ and a sufficiently small analytic neighborhood N on which the three projective roots remain distinct. Every finite root stays in a compact affine-root chart and has P' bounded away from zero; equations (10) therefore bound its source branch. A root can leave that chart only by tending to $[1 : 0]$, which forces $c \rightarrow 0$. In the $s = V/U$ chart its equation and inverse are

$$c = 2s - bs^2 + 2as^3, \text{quad}x = \frac{s}{1 - bs + 3as^2}, \text{quad}y = b - 3as,$$

together with the polynomial expression (14). At $s = c = 0$, the derivative of the first equation with respect to s is 2. The analytic implicit-function theorem therefore makes this branch regular, and it tends to $(0, b, a - 4b^2)$ rather than to infinity. Thus every source branch over N is bounded, proving $q_0 \notin S_F$. $\square$

Proposition 5.3 (normalization of the discriminant). *The morphism*

$$\nu : \mathbb{A}_{t,c}^2 \longrightarrow V(\Delta), \quad (t, c) \longmapsto (t^2 - ct^3, 4t - 3ct^2, c)$$

is the normalization. It is birational and one-to-one over the triple-root curve Γ .

Proof. The displayed point has a repeated root t , so it lies on $V(\Delta)$. In the function field of $V(\Delta)$, t satisfies the monic equation $t^2 - bt + 3a = 0$, hence is integral. Away from $3bc = 4$ it is recovered by

$$t = \frac{b - 9ac}{4 - 3bc},$$

so ν is birational. Since $\mathbb{A}^2$ is normal and the finite integral extension supplies all repeated-root parameters, it is the normalization. On Γ the unique triple root determines t , so the normalization has one point over each point of Γ . $\square$

The divisor $r = P'(t) = 0$ in the polynomial graph coordinates (t, r, c) accounts for the lost length-two part of a generic discriminant fiber. This boundary statement concerns the finite marked-root cover, not ramification of the etale affine map F .

6 The weighted polynomial construction

Let k be a field of characteristic zero. Fix a polynomial $H \in k[W]$ of degree $n \geq 3$ satisfying

$$H(0) = H(1) = 0, \text{quad}H'(0) = 0, \text{quad}H'(1) = -c \neq 0, \text{quad}\kappa := H''(1)/c \neq -2. \quad (19)$$

Thus $c \in k^\times$. Choose $b_0 \in k^\times$ and set

$$a_0 = -\frac{1 + \kappa}{2 + \kappa}. \quad (20)$$

On $\mathbb{A}_{x,y,z}^3$ define

$$v = xy, qquad S = x^2 z, qquad u = 1 + v, qquad \gamma = 1 + a_0 v + b_0 S, qquad W = u\gamma. \quad (21)$$

Write $p = H'$ and

$$q(W) = \frac{Wp(W) - H(W)}{c}. \quad (22)$$

The weighted map $G_H = (A, B, C)$ is given by

$$C = x\gamma, qquad B = \frac{c + p(W)/\gamma}{x}, qquad A = \frac{u + q(W)/\gamma^2}{x^2}. \quad (23)$$

The next proposition proves that the displayed rational expressions are actual polynomial coordinates.

For clarity, if $f \in k[x, y, z]$, write

$$\partial_x^{[1]} f = \frac{f - f|_{x=0}}{x}, qquad \partial_x^{[2]} f = \frac{f - f|_{x=0} - x(\partial f / \partial x)|_{x=0}}{x^2}.$$

These are polynomial divided-difference operators.

Proposition 6.1 (polynomiality). *Equations (23) define elements of $k[x, y, z]$. More explicitly, if $p(W) = W\tilde{p}(W)$ and $q(W) = W^2\tilde{q}(W)$, then*

$$\begin{aligned} C &= x\gamma, \\ B &= \partial_x^{[1]}(c + u\tilde{p}(u\gamma)), \\ A &= \partial_x^{[2]}(u + u^2\tilde{q}(u\gamma)). \end{aligned}$$

Proof. The conditions $p(0) = 0$ and $H(0) = H'(0) = 0$ imply respectively $W \mid p(W)$ and $W^2 \mid q(W)$. Hence

$$p(W)/\gamma = u\tilde{p}(u\gamma), qquad q(W)/\gamma^2 = u^2\tilde{q}(u\gamma)$$

are polynomials before division by powers of x .

At $x = 0$ one has $(u, \gamma, W) = (1, 1, 1)$, so the numerator of B is $c + p(1) = 0$. It is therefore divisible by x . Further,

$$q(1) = -1, qquad q'(1) = \kappa.$$

Thus the numerator $N_A = u + q(W)/\gamma^2$ vanishes at $x = 0$, and direct differentiation gives

$$\left. \frac{\partial N_A}{\partial x} \right|_{x=0} = (1 + \kappa + a_0(2 + \kappa))y = 0$$

by (20). Hence $x^2 \mid N_A$. The divided-difference formulas are therefore equal to (23) and explicitly exhibit all three coordinates as polynomials. $\square$

Proposition 6.2 (constant Jacobian). *For every H, b_0 satisfying (19),*

$$\det DG_H = b_0 c \in k^\times.$$

In particular G_H is etale and quasi-finite.

Proof. On the dense open $D(C) = D(x\gamma)$ introduce

$$s = BC = c\gamma + p(W), qquad t = cAC^2 = cW\gamma + Wp(W) - H(W). \quad (24)$$

The successive coordinate changes have Jacobian determinants

$$\begin{aligned} \det \frac{\partial(x, v, S)}{\partial(x, y, z)} &= x^3, & \det \frac{\partial(W, \gamma, C)}{\partial(x, v, S)} &= b_0\gamma^2, \\ \det \frac{\partial(s, t, C)}{\partial(W, \gamma, C)} &= -c^2\gamma, & \det \frac{\partial(s, t, C)}{\partial(A, B, C)} &= -cC^3. \end{aligned}$$

The chain rule and $C = x\gamma$ give

$$x^3 b_0 \gamma^2 (-c^2 \gamma) = (-cC^3) \det DG_H,$$

and cancellation on $D(C)$ yields $\det DG_H = b_0 c$. Both sides are polynomials, so the equality extends to all of $\mathbb{A}_k^3$. An étale morphism of finite type is quasi-finite. $\square$

The word “generic” has not been used in this section: polynomiality and the Jacobian identity hold uniformly for every H satisfying the explicit admissibility conditions (19).

7 The normalized incidence and reconstruction

Retain all hypotheses and notation from Section 6. For a target (A, B, C) define

$$E_{A,B,C}(W) = H(W) - BCW + cAC^2. \quad (25)$$

7.1 Four distinct spaces

The *raw incidence* is

$$\mathcal{I}_H = V(E_{A,B,C}(W)) \subset \mathbb{A}_{A,B,C}^3 \times \mathbb{A}_W^1. \quad (26)$$

If h_n is the leading coefficient of H , then $h_n^{-1}E$ is monic in W . Hence $\mathcal{I}_H \rightarrow \mathbb{A}^3$ is finite flat of degree n , with basis $1, W, \dots, W^{n-1}$. The projective homogenization has value $h_n W^n$ on the hyperplane at infinity, so (26) already contains all projective roots.

The polynomial E is irreducible in $k[A, B, C, W]$. Indeed, viewed as a linear polynomial in A over the UFD $k[B, C, W]$, it is primitive because cC^2 and $H(W) - BCW$ are coprime; it is irreducible over the fraction field and hence by Gauss’s lemma. Thus $\mathcal{I}_H$ is integral. We do *not* assert that it is normal.

Let

$$\nu : \tilde{\mathcal{I}}_H \longrightarrow \mathcal{I}_H \quad (27)$$

be its normalization. This is the *entire finite normal model* used below. Normality is imposed by taking normalization; it is not deduced for the raw incidence. Since a finite-type k -scheme is excellent, (27) is finite, and the composite $\tilde{\mathcal{I}}_H \rightarrow \mathbb{A}^3$ remains finite of degree n [The26a].

The *raw simple-root locus* is

$$\mathcal{I}_H^{\text{simp}} = D(E_W) \subset \mathcal{I}_H, \quad E_W = H'(W) - BC.$$

Finally, define rational functions on the common function field by

$$\gamma = \frac{BC - H'(W)}{c}, qquad x = \frac{C}{\gamma}, qquad y = \frac{W - \gamma}{C}, qquad z = \frac{\gamma - 1 - a_0(W/\gamma - 1)}{b_0 x^2}. \quad (28)$$

The *regular-reconstruction open* is the explicitly defined open

$$\mathcal{R}_H = \left\{ \xi \in \tilde{\mathcal{I}}_H : x, y, z \text{ are regular at } \xi \right\}. \quad (29)$$

It is open because the regularity domain of a rational function is open. These four objects—raw incidence, normalization, simple-root locus, and regular-reconstruction open—must not be identified without proof.

Proposition 7.1 (reconstruction over $C \neq 0$). *Over $D(C)$, normalization is unnecessary on the simple-root locus and*

$$G_H^{-1}(D(C)) \xrightarrow{\sim} \mathcal{I}_H^{\text{simp}} \cap D(C).$$

The inverse is given by (28).

Proof. On the incidence, $\gamma = -E_W/c$. Thus on $D(CE_W)$ both C and γ are units and all formulas (28) are regular. Moreover E_W is invertible there, so the raw incidence is etale over the regular target on this locus; it is normal, and the normalization is an isomorphism over it. Put $u = W/\gamma$, $v = u - 1$, and $S = (\gamma - 1 - a_0 v)/b_0$. Then $x = C/\gamma$, $y = v/x$, and $z = S/x^2$. Substitution in (23), using $E = 0$, returns (A, B, C) and $W = u\gamma$.

Conversely, at a source point with $C = x\gamma \neq 0$ one has

$$E_W(W(x, y, z)) = -c\gamma \neq 0.$$

The same identities recover (x, y, z) . They are equalities in localized coordinate rings, not merely pointwise formulas. $\square$

Theorem 7.2 (global regular-reconstruction theorem). *For every admissible H satisfying (19), the root marking*

$$\iota_0 : \mathbb{A}_{x,y,z}^3 \longrightarrow \mathcal{I}_H, qquad(x, y, z) \longmapsto (G_H(x, y, z), W(x, y, z))$$

lifts uniquely to an isomorphism

$$\iota : \mathbb{A}_{x,y,z}^3 \xrightarrow{\sim} \mathcal{R}_H.$$

Under this identification, G_H is the restriction of the finite map $\tilde{\mathcal{I}}_H \rightarrow \mathbb{A}_{A,B,C}^3$. Its generic degree is n .

Proof. The identity $E_{G_H(x,y,z)}(W(x, y, z)) = 0$ follows directly from (23) and defines ι_0 . Proposition 7.1 makes this morphism dominant and birational. Because $\mathbb{A}^3$ is normal, the universal property of normalization gives a unique lift ι to $\tilde{\mathcal{I}}_H$.

The lift is quasi-finite: each of its fibers is contained in a fiber of the etale, hence quasi-finite, map G_H . It is a quasi-finite birational morphism to the normal scheme $\tilde{\mathcal{I}}_H$, so Zariski's Main Theorem makes it an open immersion [The26c]. On its image, the rational functions (28) pull back to the source coordinates; hence the image lies in $\mathcal{R}_H$.

Conversely, wherever x, y, z in (28) are regular, they define a morphism to $\mathbb{A}^3$. The identities $G_H(x, y, z) = (A, B, C)$ and $W = (1 + xy)\gamma$, already proved in the common function field on $D(C)$, hold on that regularity neighborhood. This morphism is inverse to the open immersion, so its domain is exactly $\mathcal{R}_H$.

Finally, a generic target has $C \neq 0$ and $E_W \neq 0$ at each of the n roots of the monic degree- n polynomial E . Proposition 7.1 therefore reconstructs n source points, proving that the generic degree of G_H is n . $\square$

Because $\tilde{\mathcal{I}}_H$ is normal, regularity of a rational function is detected at height-one points. Consequently the complement is specified set-theoretically by

$$\tilde{\mathcal{I}}_H \setminus \mathcal{R}_H = \bigcup_{\substack{D \in \tilde{\mathcal{I}}_H^{(1)} \\ \min(v_D(x), v_D(y), v_D(z)) < 0}} \overline{\{D\}}. \quad (30)$$

Thus the affine source is one explicitly defined open, and its divisorial complement consists of the pole valuations (30). The argument in this paper does not enumerate these valuations for every root profile of H ; complete boundary lists are stated only when the required root hypotheses and local degree counts have been checked.

7.2 The fiber over $C = 0$

At $C = 0$, the raw incidence specializes to $H(W) = 0$. The simple-root locus is not the affine source globally: a repeated root at $W = 0$ may have normalized branches with regular reconstruction, while a simple additional root of H may have a reconstruction pole.

The distinguished root $W = 1$ is simple because $H'(1) = -c$. In the completed etale local incidence at that root,

$$W = 1 - \frac{B}{c}C + \left(A + \frac{2 + \kappa}{2c^2}B^2\right)C^2 + O(C^3). \quad (31)$$

The reconstruction then has $x \rightarrow 0$ and $y \rightarrow -(2 + \kappa)B/c$; the choice (20) cancels the linear term in $\gamma - 1 - a_0(W/\gamma - 1)$, so formally $S = O(C^2)$ and $z = S/x^2$ has no formal pole. This completed-local calculation implies algebraic regularity: for the excellent normal local ring of the normalized incidence, faithful flatness of completion gives $K \cap \hat{\mathcal{O}} = \mathcal{O}$ inside the completed fraction field.

Write $H(W) = W^{m_0}L(W)$ with $m_0 \geq 2$ and $L(0) = h_0 \neq 0$. The chart $W = CR$, after division by C^2 , has strict equation

$$C^{m_0-2}R^{m_0}L(CR) - BR + cA = 0. \quad (32)$$

For $m_0 = 2$, its generic special fiber is $h_0R^2 - BR + cA = 0$; for $m_0 \geq 3$ and $B \neq 0$, it has the generic solution $R = cA/B$. These formulas exhibit finite normalized branches but do not classify every special point or every boundary divisor over $C = 0$.

8 Irreducibility and symmetric monodromy

Let k be any field of characteristic zero and let $H \in k[W]$ have degree $n \geq 2$; admissibility at 0 and 1 is not required in this section. Put

$$\mathcal{E}(W) = H(W) - sW + t \in k[s, t, W]. \quad (33)$$

Proposition 8.1 (geometric irreducibility). *The polynomial $\mathcal{E}$ is irreducible over $\bar{k}(s, t)$ and over $k(s, t)$.*

Proof. Work first in $\bar{k}[s, W, t]$. Since $\mathcal{E}$ has degree one in t , in any factorization one factor, say A , has t -degree zero. Write the other factor as $B_0 + B_1t$. Comparison of the coefficient of t gives $AB_1 = 1$ in $\bar{k}[s, W]$, so A and B_1 are units. Thus the factorization is trivial. The leading coefficient in W is the nonzero constant leading coefficient of H , so Gauss's lemma gives irreducibility over $\bar{k}(s, t)$. The assertion over $k(s, t)$ follows. $\square$

Proposition 8.2 (discriminant normalization). *The repeated-root discriminant of (33) is an irreducible curve whose normalization is birationally parametrized by*

$$r \longmapsto (s, t) = (H'(r), rH'(r) - H(r)). \quad (34)$$

Its generic point represents a polynomial with exactly one double root and all other roots simple.

Proof. A repeated root r satisfies $\mathcal{E}(r) = \mathcal{E}'(r) = 0$, which is equivalent to (34). The two coordinate functions have pole orders $n - 1$ and n at $r = \infty$. The generic degree of the induced map from $\mathbb{P}^1$ to its image divides both pole orders, hence is one. The image is therefore irreducible and the map is birational. The morphism from $\mathbb{A}_r^1$ is finite because r is integral over $k[H'(r)]$: after division by the leading coefficient, $H'(T) - H'(r)$ is a monic equation for r over that subring. Since $\mathbb{A}^1$ is normal, it is the normalization of the affine discriminant. Since H'' is not the zero polynomial in characteristic zero, $H''(r) \neq 0$ generically; the marked repeated root is then exactly double. Birationality excludes a second generic repeated root. $\square$

Theorem 8.3 (full symmetric monodromy). *The geometric Galois group of (33) over $\bar{k}(s, t)$ and the arithmetic Galois group over $k(s, t)$ are both S_n in their natural action on the n roots.*

Proof. Let L be the splitting field over $\bar{k}(s, t)$ and let G be its Galois group. Proposition 8.1 makes the action transitive. At the generic point of the irreducible discriminant, a transverse local parameter has the form

$$\epsilon + \lambda(W - r)^2 + \text{higher terms}, \quad \lambda \neq 0.$$

Characteristic zero makes the local extension tame, and its inertia group is generated by a transposition.

Let $N \triangleleft G$ be the normal subgroup generated by all codimension-one inertia groups. The normalization of $\mathbb{A}_{\bar{k}}^2$ in L^N is unramified at every codimension-one point. Zariski–Nagata purity makes it finite étale over all of $\mathbb{A}_{\bar{k}}^2$. Every finite étale cover of affine space over an algebraically closed characteristic-zero field is trivial: descend to a finitely generated subfield, embed it in $\mathbb{C}$, and apply the Riemann existence comparison and contractibility of $\mathbb{C}^2$ [The26b, GR71]. Hence $N = G$.

Thus G is a transitive subgroup generated by transpositions. Form the graph whose edges are those transpositions; transitivity makes it connected, and the edge transpositions of a connected graph generate S_n . Therefore $G = S_n$. The arithmetic group contains the geometric group as a normal subgroup and is itself contained in S_n , so it is also S_n . $\square$

For an admissible weighted map, on $D(C)$ put $s = BC$ and $t = cAC^2$. Then $k(A, B, C) = k(s, t, C)$, a purely transcendental extension of $k(s, t)$. Theorem 8.3 therefore gives geometric and arithmetic monodromy S_n for the generic inverse cover of G_H .

9 Examples and scope of the construction

9.1 The cubic collision as three marked roots

At the collision target $(-1/4, 0, 0)$, the binary cubic is

$$Q(U, V) = -2U^2V + \frac{1}{2}V^3 = -2V(U - V/2)(U + V/2).$$

Its three simple projective roots are

$$[1 : 0], \quad [1/2 : 1], \quad [-1/2 : 1].$$

Through the two inverse charts of Section 4, these give respectively

$$(0, 0, -1/4), \quad (-1, 3/2, 13/2), \quad (1, -3/2, 13/2).$$

This recovers the collision without solving the original three coordinate equations simultaneously.

9.2 Canonical admissible polynomials

For every $d \geq 2$, let

$$H_d(W) = W^d(1 - W).$$

Then $n = d + 1$, $H_d(0) = H_d(1) = H'_d(0) = 0$, and $H'_d(1) = -1$, so $c = 1$. Moreover $H''_d(1) = -2d$, hence $\kappa = -2d \neq -2$, and

$$a_0 = \frac{1 - 2d}{2(d - 1)}.$$

For any $b_0 \in k^\times$, equations (21)–(23) therefore give a degree- $(d + 1)$ generic inverse cover with constant Jacobian b_0 and monodromy S_{d+1} . This is an application of the uniform theorems, not a bounded computation.

At $C = 0$, H_d has a root of multiplicity d at zero and a simple root at one. The only conclusions established here about this special fiber are those in Section 7: the affine source is the regular-reconstruction open of the normalized incidence, the root-one formal chart is finite, and (32) gives the generic zero-cluster chart. No exact image or complete boundary classification for these examples is asserted here.

9.3 Logical scope

The cubic source is literally the raw simple-projective-root locus. The weighted source is generally not: it is the regular-reconstruction open in the normalization of the raw affine-root incidence. The generic degree and monodromy are uniform, while exact images and special-fiber boundary strata depend on H and lie outside the theorems proved in this paper.

9.4 Comparison, verification, and provenance

The explicit map (6), its determinant, and the rational collision were publicly circulated in July 2026 in an announcement attributed in contemporaneous sources to Levent Alpöge and Fable [Alp26, Gal26, Zha26]. The direct announcement has not been archived in a form from which this paper can settle the full discovery history, and we make no priority claim.

The organization by binary cubics with one simple projective root marked was also described publicly by Andy Jiang [Jia26]. The present paper supplies the two-chart scheme-theoretic proof, including the root at infinity, and uses that formulation with attribution rather than as a claim of independent discovery. The weighted formulas in Section 6 overlap the same-day explanatory account [Gal26]; our contribution here is to state carefully the raw incidence, its normalization, and the regular-reconstruction open.

Dean Cureton independently formalized the determinant, the three rational evaluations, a determinant-one rescaling, and the complex specialization in Lean 4 [Cur26]. The pinned implementation is available at the audited commit. It does not formalize Theorems 4.4 or 7.2. We link to that implementation rather than copy it; the audited commit did not contain an explicit license file.

Exact computer-algebra scripts also check the displayed determinant, both cubic chart compositions, and representative weighted identities. Those checks are useful guards against transcription errors. They are not used as proofs of the uniform statements in H or n .

10 Further directions

The results above stop with the uniform weighted construction. Two natural questions remain. First, for a fixed admissible H , one may try to classify the pole valuations (30) and thereby determine the exact image, nonproperness set, and special fibers of G_H . Second, one may compare other one-variable inverse constructions with the weighted family under polynomial source and target automorphisms.

Subsequent work in this repository answers the degree-by-degree comparison question beyond the scope of this paper. For every generic fiber degree $N \geq 4$, there are at least $\tau(N - 1)$ pairwise stably inequivalent three-dimensional Keller maps, where τ is the positive-divisor counting function. The cancellation types are indexed by the proper divisors $r \mid N - 1$, with $m = (N - 1)/r - 1$, and are separated by their boundary ramification and nilpotency indices; a split weighted seed supplies one further reduced-boundary class. This paragraph is only a retrospective summary. The canonical statement and proof are in `papers/weighted-root-multiplicity/main.tex`; the common normalized-root-cover framework is recorded separately in `MARKED_ROOT_KELLER_MAPS.md`.

Neither direction is used in this paper. In particular, computations for special quartics, bounded-degree discriminants, finite fields, transfer blocks, and cancellation families do not prove or strengthen any theorem stated above. They may suggest conjectures or supply test cases, but the present argument depends only on the explicit cubic identities, the two projective root charts, normalization and Zariski’s Main Theorem, and the purity argument for symmetric monodromy.

A useful next foundational step would be an independent algebraic-geometry audit of Theorem 7.2, especially the passage from the formal $C = 0$ expansions to algebraic regularity and the precise divisorial structure of the complement of $\mathcal{R}_H$. Until that audit is supplied, the paper deliberately states no uniform seed-by-seed boundary theorem.

A Theorem dependencies

The following table records the logical inputs to the principal statements. “Exact algebra” means an identity proved in the indicated polynomial or localized coordinate ring; bounded computations are not dependencies.

Result	Internal dependencies	External input
Theorem 1.2	Proposition 1.1; chord identities (2)	Symmetric powers of $\mathbb{P}^1$; finite flat marked-point cover; triviality of line bundles and additive torsors on affine space
Theorem 3.1	Coordinate identity (7), chain rule, exact substitution	Jacobian criterion for etaleness over a characteristic-zero field
Theorem 4.4	Proposition 4.1; Lemmas 4.2 and 4.3; overlap (16)	None; this is an explicit two-chart coordinate certificate
Proposition 5.1	Theorem 4.4; multiplicity classification of a binary cubic; coefficient comparison for a cube	None beyond elementary algebra
Proposition 5.2	Both cubic reconstruction charts; discriminant identity	Complex implicit-function theorem and the sequential criterion for nonproperness

Result	Internal dependencies	External input
Proposition 6.1	Endpoint conditions (19); divisibility of p, q ; divided differences	None
Proposition 6.2	The four coordinate-change determinants in (24)	Etale morphisms of finite type are quasi-finite
Proposition 7.1	Incidence equation (25); explicit reconstruction (28)	None
Theorem 7.2	Propositions 6.1, 6.2, and 7.1	Finiteness and universal property of normalization; Zariski’s Main Theorem; normality criterion via height-one valuations
Proposition 8.1	Linearity in t and coefficient comparison	Gauss’s lemma
Proposition 8.2	Repeated-root equations and coprime pole orders $n - 1, n$	Standard facts on finite maps of normal projective curves
Theorem 8.3	Propositions 8.1 and 8.2; connected transposition graph	Zariski–Nagata purity; Riemann existence comparison; triviality of finite etale covers of affine space in characteristic zero

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